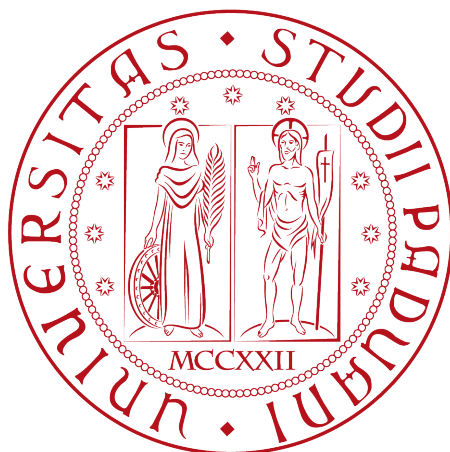


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MASTER DEGREE IN COMPUTER SCIENCE



GeoMedia: A Geographic Framework for Location-Based Content Sharing

Master's Thesis

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Summary

Over the past few decades, the use of mobile devices has increased drastically, almost replacing desktop software and web browsers. This shift has been driven not only by the widespread adoption of smartphones, but also by the availability of integrated sensors such as GPS, gyroscopes, and Bluetooth, which allows a richer and more context-aware user experiences. By leveraging these technologies, Online Social Media platforms have been able to provide more advanced and sophisticated features; consequently collecting larger amounts of metadata, interaction and daily engagement.

This thesis presents “GeoMedia”, an open-source framework designed for sharing multimedia content — such as text, images, audio, and files — anchored to real-world geographic locations. By leveraging GPS data available on mobile devices, the application provides an interactive map through which users can explore and publish digital content, creating a seamless integration between the digital and physical worlds. Moreover, the framework offers several features and cover various use cases, including restricting content visibility within customizable geographic areas, enabling time-sensitive content, applying user-based restrictions for content creation and visualization, and supporting sequential visibility between published posts. Furthermore, analyzes the motivations behind GeoMedia, its critical aspects and design considerations, describes its system architecture and implementation through a comparison of different technological paradigms. Finally, it explores potential applications and outlines possible future developments of the framework.

An previous version of this framework has been presented at “GoodIT 2025: International Conference on Information Technology for Social Good” (September 3 - 5, 2025 - Antwerp, Belgium) [1]

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Chapter 1

Introduction

1.1 Overview

Technology has always influenced society, from the first introduction to the Personal Computer to the Internet, changing the world year after year, from personal life to workspaces, people are surrounded by software on their computers and mobile devices. Over the last decades, the investments in the development of technological innovations, both in hardware and software, as increased exponentially, shaping economical and social impact by integrating the latest features into everyday life. Smartphones and tablets have been the core engine that completely changed daily habits and digital usage; their wide adoption has reached almost three quarters of global population [24, 22], making users abandon traditional computers, not just for web browsing but more especially for social media[21] and instant messaging platform[23], that changed the design and development of applications into a ‘mobile first’ logic.

Compared to computers, mobile devices offer distinct functionalities and strengths that have facilitated their widespread adoption. These can be categorized according to the fundamental logic underlying any technology:

1. **Hardware:** the increase in computational performance[15], battery life[4], fast connectivity, and the integration of several built-in sensors[14] such as gyroscopes, GPS, NFC, light sensors, cameras, and vibration motors. These enable applications to interact with different fields such as health[2], education[17], and, through motion-sensors or cameras, an interaction with physical world, creating more immersive user experience.
2. **Software:** the development and presence of mobile applications has played a crucial role. In fact, platforms such as ‘Windows Phone’ were eventually abandoned and discontinued due to the lack of applications [3, 25] compared to operating systems like ‘iOS’[7] and ‘Android’[8].

In this way, smartphones apps have become a key integration with daily life and physical world, for example, when planning a trip or just driving, smartphones, integrated with cars assists users via app like ‘Google Maps’[10] or ‘Waze’[5], providing not just directions but also real-time traffic updates, read events or information about available parking. This makes the surrounding environment enriched with digital details. Another example is ‘Google Lens’[9], which can identify objects from a photograph, or ‘Shazam’[20] that through devices’ microphone can recognize a playing song. Even in workplaces or public areas, technology integrations can be present, from the restaurant that provide a digital menu by a tablet, to customers checking the nutrition macros into a grocery store by scanning the barcode on the pack, to museums by scanning a QR Code enabling audio-guided tours or unlock Augmented Reality functionalities.

1.2 Social networks

Alongside the widespread adoption and continuous advancement of smartphones technologies and other mobile devices, social networks, have emerged as one of the most influential and important sectors of the technology market. Today, social media platforms rank among the firsts widely used applications worldwide [18, 6], while the companies that operate them become some of the most bigger actors in global financial markets [19, 12, 11].

Social media can be divided into different subgroup, like instant-messaging or online-social-network, but despite their nature they all provide some key points like instant-time messaging capabilities, both one-to-ton as one-to-many communications till to be integrated in

Platform like WhatsApp, Telegram, Instagram, and Facebook have totally changed the way people interact among themself digitally, by providing instant-time messaging capabilities, both one-to-one as one-to-many communication and also enabling users to share any kind of multimedia (including photos, videos, audio recordings and other files) till to be integrated into everyday communication practices across social, professional, and public environments.

The content shared, usually referred as ‘post‘ crate a starting point for interactions between users, throught comments and reactions (i.e. ‘like’,‘dislike’), thus to digitally replicate human behavior during a dialogs and interactions

In addition to messaging capabilities, these platforms allow users to create and share a wide range of multimedia content, including images, videos, audio recordings, and documents. Such content, commonly referred to as a *post*, serves as the basis for user engagement through comments, reactions, and content redistribution mechanisms. These features digitally replicate several aspects of human social interaction, fostering the creation of large online communities and enabling information to spread rapidly across extensive networks of users.

Despite the significant benefits offered by social media platforms, including enhanced connectivity, community building, and information sharing, their widespread adoption has also introduced a number of societal and ethical challenges. Concerns regarding user privacy, the collection and monetization of personal data, the dissemination of misinformation and fake news, cyberbullying, stalking, online harassment, and the circulation of illegal or harmful content have become increasingly relevant topics of discussion among researchers, policymakers, and the general public.

To address these issues, social media platforms have implemented various moderation and safety mechanisms, including automated content analysis systems, community guidelines, reporting tools, and human review processes. While these measures contribute to reducing the spread of harmful content, they are not always fully effective due to the scale and complexity of modern social networks. Consequently, user reporting and human moderation continue to play a crucial role in identifying, evaluating, and removing content that violates platform policies or legal regulations.

Chapter 2

Related works

SOCIAL MEDIA

- snapchat - instagram - google maps / trip advisor - foursquare - jagat - auglinn - wallame??

2.1 Instagram and Snapchat

2.2 Google Maps

2.3 Strava

2.4 WallaMe

Chapter 3

GeoMedia

3.1 Idea

3.2 Functionalities

Chapter 4

Conclusions

- 4.1 Consuntivo finale
- 4.2 Raggiungimento degli obiettivi
- 4.3 Conoscenze acquisite
- 4.4 Valutazione personale

“Life is really simple, but we insist on making it complicated”

— Confucius

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Padua, May 2026

Alberto Morini

Appendix A

Appendice A

Citazione

Autore della citazione

Acronimi e abbreviazioni

API [Application Program Interface](#). [9](#)

UML [Unified Modeling Language](#). [9](#)

Glossario

API in informatica con il termine *Application Programming Interface API* (ing. interfaccia di programmazione di un'applicazione) si indica ogni insieme di procedure disponibili al programmatore, di solito raggruppate a formare un set di strumenti specifici per l'espletamento di un determinato compito all'interno di un certo programma. La finalità è ottenere un'astrazione, di solito tra l'hardware e il programmatore o tra software a basso e quello ad alto livello semplificando così il lavoro di programmazione. [8](#)

UML in ingegneria del software *UML, Unified Modeling Language* (ing. linguaggio di modellazione unificato) è un linguaggio di modellazione e specifica basato sul paradigma object-oriented. L'*UML* svolge un'importantissima funzione di “lingua franca” nella comunità della progettazione e programmazione a oggetti. Gran parte della letteratura di settore usa tale linguaggio per descrivere soluzioni analitiche e progettuali in modo sintetico e comprensibile a un vasto pubblico. [8](#)

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